

The result above deals only with the case of two subspaces. When asking about a possible direct sum with more than two subspaces, it is not enough to test that each pair of the subspaces intersect only at 0. To see this, consider Example 1.44. In that nonexample of a direct sum, we have $V_1 \cap V_2 = V_1 \cap V_3 = V_2 \cap V_3 = \{0\}$.

Sums of subspaces are analogous to unions of subsets. Similarly, direct sums of subspaces are analogous to disjoint unions of subsets. No two subspaces of a vector space can be disjoint, because both contain 0. So disjointness is replaced, at least in the case of two subspaces, with the requirement that the intersection equal $\{0\}$.

Exercises 1C

- 1 For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.
 - (a) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
 - (b) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
 - (c) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1x_2x_3 = 0\}$
 - (d) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}$
- 2 Verify all assertions about subspaces in Example 1.35.
- 3 Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.
- 4 Suppose $b \in \mathbf{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbf{R}^{[0,1]}$ if and only if $b = 0$.
- 5 Is $\mathbf{R}^2$ a subspace of the complex vector space $\mathbf{C}^2$?
- 6
 - (a) Is $\{(a, b, c) \in \mathbf{R}^3 : a^3 = b^3\}$ a subspace of $\mathbf{R}^3$?
 - (b) Is $\{(a, b, c) \in \mathbf{C}^3 : a^3 = b^3\}$ a subspace of $\mathbf{C}^3$?
- 7 Prove or give a counterexample: If U is a nonempty subset of $\mathbf{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbf{R}^2$.
- 8 Give an example of a nonempty subset U of $\mathbf{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbf{R}^2$.
- 9 A function $f: \mathbf{R} \rightarrow \mathbf{R}$ is called *periodic* if there exists a positive number p such that $f(x) = f(x + p)$ for all $x \in \mathbf{R}$. Is the set of periodic functions from $\mathbf{R}$ to $\mathbf{R}$ a subspace of $\mathbf{R}^{\mathbf{R}}$? Explain.
- 10 Suppose V_1 and V_2 are subspaces of V . Prove that the intersection $V_1 \cap V_2$ is a subspace of V .

- 11** Prove that the intersection of every collection of subspaces of V is a subspace of V .
- 12** Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.
- 13** Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

This exercise is surprisingly harder than Exercise 12, possibly because this exercise is not true if we replace $\mathbf{F}$ with a field containing only two elements.

- 14** Suppose

$$U = \{(x, -x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\}.$$

Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

- 15** Suppose U is a subspace of V . What is $U + U$?
- 16** Is the operation of addition on the subspaces of V commutative? In other words, if U and W are subspaces of V , is $U + W = W + U$?
- 17** Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V , is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

- 18** Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?
- 19** Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

- 20** Suppose

$$U = \{(x, x, y, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\}.$$

Find a subspace W of $\mathbf{F}^4$ such that $\mathbf{F}^4 = U \oplus W$.

- 21** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find a subspace W of $\mathbf{F}^5$ such that $\mathbf{F}^5 = U \oplus W$.

- 22** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbf{F}^5$, none of which equals $\{0\}$, such that $\mathbf{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

- 23** Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

Hint: When trying to discover whether a conjecture in linear algebra is true or false, it is often useful to start by experimenting in $\mathbf{F}^2$.

- 24** A function $f: \mathbf{R} \rightarrow \mathbf{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbf{R}$. A function $f: \mathbf{R} \rightarrow \mathbf{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbf{R}$. Let V_e denote the set of real-valued even functions on $\mathbf{R}$ and let V_o denote the set of real-valued odd functions on $\mathbf{R}$. Show that $\mathbf{R}^{\mathbf{R}} = V_e \oplus V_o$.

Exercises 2A

- 1 Find a list of four distinct vectors in $\mathbf{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}.$$

- 2 Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

- 3 Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$.

- 4 (a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.
(b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

- 5 Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbf{R}^3$.

- 6 Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbf{F}^3$ if and only if $c = 8$.
7 (a) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{R}$, then the list $1 + i, 1 - i$ is linearly independent.
(b) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{C}$, then the list $1 + i, 1 - i$ is linearly dependent.

- 8 Suppose v_1, v_2, v_3, v_4 is linearly independent in V . Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

- 9 Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V , then

$$5v_1 - 4v_2, v_2, v_3, \dots, v_m$$

is linearly independent.

- 10** Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbf{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \dots, \lambda v_m$ is linearly independent.
- 11** Prove or give a counterexample: If $v_1, \dots, v_m$ and $w_1, \dots, w_m$ are linearly independent lists of vectors in V , then the list $v_1 + w_1, \dots, v_m + w_m$ is linearly independent.
- 12** Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \dots, v_m + w$ is linearly dependent, then $w \in \text{span}(v_1, \dots, v_m)$.
- 13** Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Show that
- $$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m).$$
- 14** Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that the list $v_1, \dots, v_m$ is linearly independent if and only if the list $w_1, \dots, w_m$ is linearly independent.

- 15** Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$.
- 16** Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbf{F})$.
- 17** Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .
- 18** Prove that $\mathbf{F}^\infty$ is infinite-dimensional.
- 19** Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.
- 20** Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \dots, m\}$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.